

Weekly Research Report

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1 Objectives This Week

- Review about Floquet Port Theory and Practice

2 Tasks Completed

- Review Floquet Port Simulation and Fix Simulation Mistake

3 Results and Observations

- Realize previous simulation is wrong setup for design Transmitarray

4 Challenges

- In-depth review and understanding of the theoretical foundations behind metasurface array configurations.
- Investigation of alternative single-unit cell structures to evaluate phase control performance and transmission efficiency.

5 Plan for Next Week

- Perform Floquet port simulation of the unit cell to analyze periodic boundary conditions and extract phase/transmission characteristics.
- Develop an automated workflow for iterative antenna array design and parameter sweeping.
- Compare far-field patterns at various frequencies (e.g., 35, 38, 40 GHz).

References